

How the number is made

Everything that goes into deciding a child has a spike, from a single click on a screen to a candidate a guide is allowed to promote. Every constant, definition and formula below is read from the running engine, and every worked figure was computed by it.

BETA-BERNOULLI, ONE CELL AT A TIME ENGINE · @GT100K/INTEREST-INFERENCE SIGNALS · @GT100K/SIGNAL-PIPELINE

Six stages, and the last one is a person.

01 RAW Interaction A child touches something. Who, what, when, and whether we nudged them there.	02 CLASSIFY Event kind One of eleven kinds. Three of them can never score.
03 WEIGHT Evidence for and against Each kind carries a fixed weight, aged by how long ago it happened.	04 FOLD α and β per cell A cell is one subject crossed with one way of working.
05 POSTERIOR Mean, spread, lower bound We report the pessimistic edge, not the average.	06 GATE Candidate Four conditions at once, then a named human decides.

A cell is the unit of belief. Not a subject, and not a child. It is one (*domain* $\times$ *mode*) pair, for example `MUSIC & SOUND  $\times$  BUILD` or `MATH & PUZZLES  $\times$  INVESTIGATE` . The same child holds a separate belief for every cell they have touched, which is how the system can say a child is drawn to the subject, the way of working, or both.

What each event kind actually means.

These are the exact rules, not paraphrases. There is no *voluntary return* event kind, because the concept was too vague to score: it once covered both "reopened it thirty seconds later" and "came back on day four unprompted", which are not the same claim. The two kinds below replaced it. **voluntary** survives only as a flag on the raw interaction, meaning "we did not nudge them".

KIND	FIRES WHEN	WHAT IT ASSERTS	WEIGHT
cross_day_return	The latest engagement of this same cell by this same child strictly before this one, prompted or not, falls on an earlier UTC calendar day , and the child was not prompted here. Only the single most recent predecessor is consulted, so a same-day re-entry masks an earlier-day one.	They came back on their own after a night away. THE SIGNAL	α +1.0
same_day_engagement	Everything else that is an engagement: a first-ever touch, a reopen inside one session, or a re-entry in a <i>different</i> session on the <i>same</i> calendar day.	Nothing about delayed return. A return hours later has not survived a night away.	0
prompted_return	The interaction arrived with <code>returnState === "prompted"</code> . The system surfaced, nudged or recommended the child into it.	That we are good at recommending. Shown to a guide as disconfirming context.	0
unrequired_revision chosen_challenge failure_recovery self_authored_scope	The occurrence exhibited one of these four depth families. Presence is what counts; magnitude is never read as a strength, only tested for being positive.	They did something harder than they had to. This is the within-day discriminator that time was supposed to be.	α +0.5
artifact_competence	The fifth depth family. Emitted, stored, and read by the specialization planner.	How good the made thing is, which is a different question. A child can do competent work in something that bores them.	0
external_report scope: broad	An <code>external_report</code> event carries <code>reportScope: "broad"</code> , meaning the interest reached <i>the topic itself</i> rather than the materials the child was handed. Attributed to a named adult, and collected in a delayed interview.	Weak corroboration. Deliberately below a depth signal, and it never marks a day.	α +0.25
external_report scope: focused	The same report, coded as having stayed bound to the materials.	Evidence about the wrapper, not the domain. Scoring it would throw away the discrimination the coding exists to make.	0
skip	The cell was on offer in a session, the child did not take it, and the child has engaged this cell at some point in the processed history.	They know what it is and passed anyway.	β +0.5 ÷ choice set
decline	The cell was on offer, not taken, and the child has never engaged it at all. Disjoint from skip by construction; one cell in one session yields at most one of the two.	Much weaker. You cannot tire of something you never tried.	β +0.15 ÷ choice set
novelty a flag, not a kind	A boolean set on <i>any</i> event, checked before the kind is. True within 3 days of that child's first exposure to the cell, and first exposure itself is novelty. First exposure means first engagement for engagement events, and first engagement <i>or surfacing</i> for skips and declines. An unparseable timestamp is treated as novelty.	Triggered situational interest. Excluded entirely, in both directions.	0

Two rules keep the negatives honest. A session in which the child neither engaged nor opened anything produces no skips and no declines, because a child who could not find a way in is not a child who rejected everything. And merely opening something counts as taking it, not as passing it over, so a child who glances and leaves does not earn a decrement against the one cell they demonstrably chose to look at.

One choice moment is worth one skip, however many things were on screen.

choiceSetSize is the number of cells that were offered, *not* taken, and past their novelty window. Each one's weight is divided by it, so the six alternatives you passed over to pick a seventh share a single skip's worth of disconfirming mass between them rather than each taking a full one. Without this, a child who returns to their favourite cell every session accrues a full decrement against every other known cell every session, and breadth of discovery would suppress the very spike the engine exists to find.

CHOICESETSIZE (cells offered, not taken, past novelty)	ONE SKIP ADDS TO B	ONE DECLINE ADDS TO B
1 or absent, or zero	0.5000	0.1500
2	0.2500	0.0750
3	0.1667	0.0500
5	0.1000	0.0300
7	0.0714	0.0214
10	0.0500	0.0150

Figures above are before recency decay. This single rule is the difference between case B passing and case B failing, which is the callout under the worked examples.

A coin flip, nudged by circumstance.

Before a child does anything, every cell sits at even odds. Three facts about the child's world can tilt the starting point, and only upward.

$\alpha_{\theta} =$

1

+

0.5

if the domain exists in their environment

+

0.5

×

masteryTilt

// how they already do in it at school, [0,1]

+

0.5

×

discretionaryTilt

// free-choice time they already give it, [0,1]

$\beta_{\theta} =$

1

// nothing tilts the against side. There is no prior reason to disbelieve.

SITUATION	A _θ	B _θ	STARTING MEAN
Nothing known about the child	1.0	1.0	0.500
Domain exists in their environment	1.5	1.0	0.600
Environment, top mastery, top discretionary time	2.5	1.0	0.714

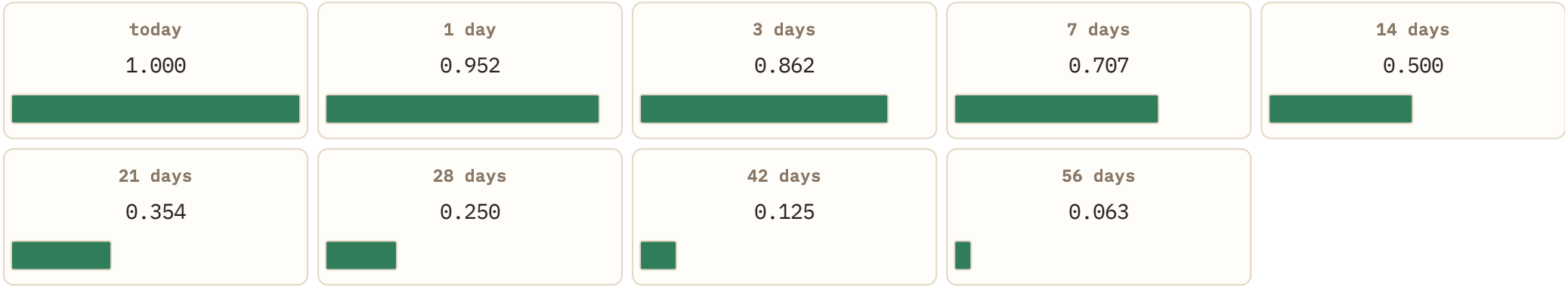
A prior can never make a cell confident. The sufficiency gates in stage 06 count only evidence recorded after the prior, and the day gate counts only days the child acted, so the most favourable prior in the world still leaves a cell uncertain until the child has actually done something on separate days. This is also why morning academic data can seed an afternoon prior without ever gating what a child may explore.

Everything halves every fourteen days.

$r(t) =$

0.5 ^ (age in days ÷ 14)

// applied to every weight before it reaches α or β



Two masses, because decay answers the wrong question twice. Decay is geometric, so a steady cadence converges: fortnightly returns ceiling at a decayed mass of 2.0 against a sufficiency floor of 6, meaning a child who came back every other week for years could never be confident at any number of visits. So the engine keeps both.

A AND B carry the decayed weights and answer *what do we believe now*. **OBSERVEDMASS** sums the same weights **undecayed** and answers *have we seen enough to say anything at all*. Observing does not un-happen because time passed.

The whole posterior, in seven lines.

```
α =      α₀ + Σ ( weight × roleScale × r(t) )  // returns and depth
      + Σ ( 0.25 × r(t) )  // broad reports; they carry no role
β =      β₀ + Σ ( weight × r(t) ÷ max(1, choiceSetSize) )  // skips and declines

n =      α + β
mean =   α ÷ n
variance = (α × β) ÷ ( n × n × (n + 1) )  // Beta distribution variance
sd =     √variance
lowerBound = max( 0 , mean - 1.0 × sd )  // the number everything downstream reads
```

Why the lower bound and not the mean

A cell with two events and a cell with twenty can share a mean of 0.85. Subtracting a standard deviation separates them, because the thin one has a wider spread and loses more. The system under-claims by construction, and the penalty falls hardest on exactly the evidence that deserves least trust.

What it is not

This is a deliberately cheap conservative point estimate, not a Beta credible interval. A proper quantile is the obvious upgrade and is not built. Said plainly here so nobody has to discover it under questioning.

roleScale is the only remaining multiplier, and it is 0.5 for the secondary reading of a two-mode action, where the mode was afforded rather than directly acted. It exists as a named engine constant rather than a number on the event, because the field it replaced was free-numeric and an emitter could have filled it with active time.

Four conditions, all at once, then a human.

A cell becomes a promotable candidate only when every one of these holds. Three of them govern whether the engine is allowed to have an opinion; the fourth is the opinion itself.

observedMass ≥ 6
Undecayed. Roughly a week of coming back, because at most one return banks per day.

distinctDays ≥ 2
Distinct UTC days that *moved* the belief. A count can be run up in one sitting; a calendar cannot.

2 × sd ≤ 0.35
The band has to be narrow enough to be worth saying out loud.

lowerBound ≥ 0.6
A cell with no evidence sits at 0.5. This is meaningfully better than knowing nothing.

Only three kinds of thing can mark a day: a cross-day return, a scoring depth family, or a skip or decline. A prompted return, a same-day engagement, a novelty event and an adult report all leave **distinctDays** where they found it.

The adult report is the pointed one. It *does* count toward the mass floor, and it never marks a day, so no quantity of parent reports can make a cell confident on its own. A child must have acted on at least two separate days before anyone is allowed an opinion.

After the gates

Passing cells are ranked by lower bound and the top **3** are surfaced. Each is attributed to **DOMAIN**, **STYLE** or **MIXED** by comparing the evidence-weighted marginal across that subject against the marginal across that way of working, with a margin of **0.1**. None of this promotes anything. Promotion is a separate human action behind three further checks.

Six returns, and how little it takes to change the verdict.

Every figure below was computed by running the engine, not by hand. All four start from a flat prior of $\alpha_0 = \beta_0 = 1$. A and B share the same six returns; B survives ten skips on top of them. C and D then fail on *different* gates while scoring well above threshold, which is the point: the four conditions are not redundant, and a high number alone is never enough.

A · Clean

Six unprompted cross-day returns at 1, 3, 6, 9, 12 and 15 days ago. No skips.

α	5.2250
β	1.0000
n	6.2250
mean	0.8394
sd	0.1366
$2 \cdot \text{sd} \leq .35$	0.2732
mass ≥ 6	6.000
days ≥ 2	6
LB ≥ 0.6	0.7027

CANDIDATE

B · Same six, plus ten skips

Each skip taken from a choice set of seven, so each contributes $0.5 \div 7$.

α	5.2250
β	1.5230
n	6.7480
mean	0.7743
sd	0.1502
$2 \cdot \text{sd} \leq .35$	0.3004
mass ≥ 6	6.714
days ≥ 2	13
LB ≥ 0.6	0.6241

CANDIDATE, NARROWLY

C · All in one day

Six returns forced onto one calendar day. The pipeline cannot produce this; it isolates the day gate.

α	6.7102
β	1.0000
n	7.7102
mean	0.8703
sd	0.1138
$2 \cdot \text{sd} \leq .35$	0.2277
mass ≥ 6	6.000
days ≥ 2	1
LB ≥ 0.6	0.7565

FAILS THE DAY GATE

D · Three returns, three days

The score is high enough. There simply is not enough of it yet.

α	3.7194
β	1.0000
n	4.7194
mean	0.7881
sd	0.1709
$2 \cdot \text{sd} \leq .35$	0.3417
mass ≥ 6	3.000
days ≥ 2	3
LB ≥ 0.6	0.6172

FAILS THE MASS GATE

Case B is the one worth arguing about. Ten skips barely move it, because each is a single share of a seven-way choice. Remove the choice-set normalisation and the identical behaviour scores **$\beta = 4.66$** , mean 0.5285, **lower bound 0.3772**, and the child loses their spike for the crime of looking at six other things. That is the whole argument for the divisor, in one number.

Case A, event by event

EVENT	AGE	R (T)	ADDS TO A	RUNNING A	UNDECAYED MASS
prior	—	—	—	1.0000	0.0
cross_day_return	1 d	0.9517	+0.9517	1.9517	1.0
cross_day_return	3 d	0.8620	+0.8620	2.8137	2.0
cross_day_return	6 d	0.7430	+0.7430	3.5567	3.0
cross_day_return	9 d	0.6404	+0.6404	4.1971	4.0
cross_day_return	12 d	0.5520	+0.5520	4.7492	5.0
cross_day_return	15 d	0.4758	+0.4758	5.2250	6.0
n = 6.2250 · mean = 0.8394 · sd = 0.1366				LB 0.7027	6.0 ✓

08 · WHAT NEVER REACHES THE NUMBER

The exclusions are the design.

Most of what a system like this could measure is noise, and some of it is actively misleading. Each of these is recorded, most are shown to a guide, and none of them can move a belief.

EXCLUDED	WHY	HOW THE EXCLUSION IS ENFORCED
Time on task	Dwell is non-monotonic in interest. Stronger performers stay as progress rises, weaker ones stay with the most familiar option. Same number, opposite meaning. At ages 7 to 8, in-session engagement failed to separate an intrinsically motivating task from a boring one; only a delayed test did.	Captured as one of four ordinal buckets on the raw interaction, and deliberately absent from the event type the belief math sees. A closed enum cannot be multiplied into α .
Attempts and success rate	A child who loves a thing they are bad at must read exactly as interested as one who finds it easy.	Lives on the raw interaction for the wellbeing engine. Never copied onto the event type.
Work quality	Competent work in a boring subject is common, and so is poor work in a beloved one.	<code>artifact_competence</code> is filtered out of the interest-scoring depth set by name.
Position on screen	People pick the top of a list far more often than the bottom whatever is in it.	Recorded at surfacing time and otherwise unused, because the size of the effect cannot be known before the surface exists. The wall also randomises order per session.
Anything the child says they like	Self-report predicts behaviour only modestly, and at these ages almost every child says yes to almost every question.	There is no such input.
Any reward for exploring	Attach one and every child explores, so the signal measures prize-chasing. For children who already had the interest, an expected tangible reward can crowd it out.	No streaks, badges, points, notifications or unlocks exist on the discovery surface.

The structural guarantee. Duration, attempts and dwell all live on the raw interaction type. The belief math only ever receives the event type. Keeping these fields off that type makes it impossible for performance or time to become interest evidence, which is a stronger promise than a comment asking people not to do it.

The whole model is twenty numbers.

Weights

A_RETURN	1.0	cross-day return
A_DEPTH	0.5	scoring depth family
A_REPORT_BROAD	0.25	adult report, broad
B_SKIP	0.5	known, passed over
B_DECLINED	0.15	never tried, passed over
A_SECONDARY	0.5	afforded, not acted

Prior

ALPHA0 / BETA0	1 / 1	even odds
W_ENV	0.5	in their environment
W_MASTERY	0.5	school achievement tilt
W_XP	0.5	discretionary time tilt

Gates

MIN_EVIDENCE_MASS	6	undecayed, raised from 3
MIN_DISTINCT_DAYS	2	UTC days that scored
MAX_CI_WIDTH	0.35	on 2·sd
SPIKE_THRESHOLD	0.6	on the lower bound
K_LCB	1.0	sd multiplier

Time and output

HALFLIFE_DAYS	14	recency half-life
noveltyWindowDays	3	excluded after first exposure
MAX_CANDIDATES	3	surfaced at once
ATTR_MARGIN	0.1	domain versus style

These are priors, not fitted parameters. They were chosen from the literature and from arguments recorded next to each one in the source. Nobody has calibrated them against a real cohort, because no real child has used this. The first cohort recalibrates all of them, and the two most likely to move are the mass floor and the spike threshold.

The engine never promotes anything.

A candidate is the most the arithmetic is allowed to produce. Turning one into a specialization a child actually pursues requires three further checks and a named adult who clicks.

Survived a gap

Two *consecutive* returns at least 14 days apart. A real quiet spell they came back from.

Durable

Two review cycles spanning at least 56 days.

Made something

An artifact that survived a setback.

A person

Who also confirms the child wants it, and whose name is on it.

Nothing demotes on silence. A quiet month is not evidence of not caring. What moves a hypothesis down is contesting evidence, and unresolved family pressure holds promotion for that child entirely until a guide reviews it.

Constants and rules read from @gt100k/interest-inference (model.ts, fold.ts, posterior.ts, aggregate.ts) and @gt100k/signal-pipeline (returns.ts, skips.ts, novelty.ts, model.ts). All worked figures computed by executing the engine, not derived by hand.

Every child, name and number in the product screenshots accompanying this document is synthetic. No real child has used any of this.